

Exp :- 3

Charge 1 = 3×10^{-6}

Charge 2 = 5×10^{-6}

$r = 3 \text{ cm} = 0.03 \text{ m}$

Current 1 = 2 A

Current 2 = 3 A

$L = 1 \text{ mm} = 0.001 \text{ m}$

$D = 10 \text{ m}$

i) Force B/w Two electric Charge

Formula: $\frac{1}{4\pi\epsilon_0} \cdot \frac{q_1 \cdot q_2}{r^2}$

given, $q_1 = 3 \times 10^{-6} \text{ C}$

$q_2 = 5 \times 10^{-6} \text{ C}$

$r = 0.03 \text{ m}$

$\epsilon_0 = 8.854 \times 10^{-12} \text{ F/m}$

Coulomb const $k = \frac{1}{4\pi\epsilon_0} = 8.99 \times 10^9 \text{ N.m}^2/\text{C}^2$

calculation :-

$$F = (8.99 \times 10^9) \times \frac{(3 \times 10^{-6})(5 \times 10^{-6})}{(0.03)^2}$$

$$F = (8.99 \times 10^9) \times \frac{15 \times 10^{-12}}{9 \times 10^{-4}}$$

$$F = (8.99 \times 10^9) \times (1.67 \times 10^{-8})$$

$$F = 150 \text{ N} //$$

$$\therefore 149.8 \text{ N} \approx 150 \text{ N}$$

ii) Force between two Parallel Condition

Formula:-

$$F = \frac{\mu_0 I_1 I_2 L}{2\pi d}$$

$$\mu_0 = 4\pi \times 10^{-7} \text{ H/m}$$

$$F = \frac{(4\pi \times 10^{-7}) (2)(2)(0.01)}{2\pi(10)} \Rightarrow F = \frac{24\pi \times 10^{-10}}{20\pi}$$

$$F = 1.2 \times 10^{-10} \text{ N}$$

